

A Finite-Second Obstruction to Universal Operators with No Unimodular Eigenvalues

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Source and Target Signal

The source paper is Sophie Grivaux, *Some new examples of universal hypercyclic operators in the sense of Glasner and Weiss*, arXiv:1410.2957, Trans. Amer. Math. Soc. 369 (2017), 7589–7629.

In Section 5, Grivaux asks whether there exist universal, or 2-universal, operators admitting no unimodular eigenvalue. She then points to the Kalish-type operator on

$$C_0([0, 2\pi]) = \{f \in C([0, 2\pi]) : f(0) = 0\}$$

as a potential example: it has no unimodular eigenvalue and nevertheless admits a strongly mixing invariant Gaussian measure with full support. The same source asks whether this particular operator is universal, or 2-universal, for invertible ergodic systems.

This brings us back to questions about the existence of unimodular eigenvectors for universal operators.

Question 5.8. — Does there exist universal (or 2-universal) operators admitting no unimodular eigenvalue? What about universal operators on a Hilbert space?

The second half of this question seems hard, again because we do not know whether a frequently hypercyclic operator on a Hilbert space necessarily has some unimodular eigenvalue. The first half of Question 5.8 may be more tractable, and a potential example would be the Kalish-type operator A of Example 4.4 acting on the space $\mathcal{C}_0([0, 2\pi])$ of continuous functions on $[0, 2\pi]$ vanishing at 0. It is proved in [3] that although this operator has no unimodular eigenvalue, it admits a Gaussian invariant measure with full support with respect to which it is strongly mixing. We have seen that A acting on $L^2(\mathbb{T})$ is universal for invertible ergodic systems, so one may naturally wonder about the following:

Question 5.9. — Let A be the bounded operator on $\mathcal{C}_0([0, 2\pi])$ defined by setting, for every $f \in \mathcal{C}_0([0, 2\pi])$ and every $\theta \in [0, 2\pi]$,

$$Af(\theta) = e^{i\theta}f(\theta) - \int_0^\theta ie^{it}f(e^{it})dt.$$

Is A a universal (or 2-universal) operator for invertible ergodic systems on $\mathcal{C}_0([0, 2\pi])$?

A positive answer to Question 5.9 cannot be obtained via an application of Theorem 1.3, or any variant of it, since we have seen that assumption (c) of Theorem 1.3 for instance implies that A admits a continuous unimodular eigenvectorfield E which is such that $\text{span}[E(\lambda); \lambda \in \mathbb{T}] = Z$. So we finish the paper with this last question:

Question 5.10. — Does there exist any universal operator for invertible ergodic systems which does not satisfy the assumptions of Theorem 1.3?

References

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- [2] F. BAYART, S. GRIVAUX, Frequently hypercyclic operators, *Trans. Amer. Math. Soc.* **358** (2006), no. 11, 5083–5117.
- [3] F. BAYART, S. GRIVAUX, Invariant Gaussian measures for operators on Banach spaces and linear dynamics, *Proc. Lond. Math. Soc.* **94** (2007), 181–210.
- [4] F. BAYART, É. MATHERON, Dynamics of linear operators, Cambridge Tracts in Mathematics **179**, Cambridge University Press (2009).
- [5] F. BAYART, É. MATHERON, Mixing operators and small subsets of the circle, *J. Reine Angew. Math.* (to appear), available at <http://arxiv.org/abs/1112.1289>.

Open-Problem Transcription

Question 5.8 in the source asks:

Does there exist universal (or 2-universal) operators admitting no unimodular eigenvalue? What about universal operators on a Hilbert space?

Question 5.9 asks whether the Kalish-type operator A on $\mathcal{C}_0([0, 2\pi])$ is universal, or 2-universal, for invertible ergodic systems. The source notes that this operator has no unimodular eigenvalue.

Result Claimed Here

The finite-second part has a general negative answer. If an operator is 2-universal for invertible ergodic systems, then it must have at least one unimodular eigenvalue. Consequently, the no-eigenvalue Kalish-type operator in Question 5.9 is not 2-universal for invertible ergodic systems.

This is a partial result for the source questions: it settles the 2-universal no-eigenvalue clause, but it does not decide ordinary universality without the finite-second moment requirement.

Proof Intuition

Test 2-universality on a nontrivial pure point ergodic system, for instance an irrational circle rotation. A finite-second invariant measure on a Banach space lets us form the Hilbert space generated by the continuous linear observables. If the modeled system has pure point spectrum, this Hilbert subspace contains a nonzero Koopman eigenfunction. The covariance/barycenter map sends that Koopman eigenfunction back to a genuine vector of the Banach space. The intertwining relation converts the Koopman eigenvalue into an eigenvalue of the original operator.

Thus the moment condition gives a bridge from measurable eigenfunctions of the modeled rotation to actual linear eigenvectors of the Banach operator. An operator with no unimodular eigenvectors cannot cross that bridge.

Theorem 1. *Let X be a complex separable Banach space and let $A \in \mathcal{B}(X)$. Suppose that there is an A -invariant Borel probability measure ν on X with full support and*

$$\int_X \|x\|^2 d\nu(x) < \infty,$$

such that the measure-preserving system $(X, \mathcal{B}_X, \nu; A)$ is isomorphic to a nontrivial invertible ergodic system with pure point Koopman spectrum. Then A has a unimodular eigenvalue.

Proof. Let U denote the Koopman operator on $L^2(X, \nu)$, $Uf = f \circ A$. Since the system is isomorphic to an invertible pure point system, U is a unitary operator with pure point spectrum.

First consider the barycenter

$$b = \int_X x d\nu(x),$$

which exists as a Bochner integral because ν has a finite second moment. Invariance gives $Ab = b$. If $b \neq 0$, then 1 is an eigenvalue of A , and we are done. Hence assume $b = 0$.

For $x^* \in X^*$, write $j(x^*)$ for the linear observable

$$j(x^*)(x) = x^*(x) \quad (x \in X).$$

Because $b = 0$, these observables have mean zero. Let

$$H = \overline{j(X^*)}^{L^2(\nu)}.$$

The subspace H is nonzero: if every continuous linear functional vanished in $L^2(\nu)$, then the full support of ν would be contained in the intersection of the kernels of a separating family of functionals, hence in $\{0\}$, impossible unless $X = \{0\}$; and in the trivial space there is no full-support model of a nontrivial system. Moreover, H is invariant under U , since

$$Uj(x^*) = j(A^*x^*).$$

As U has pure point spectrum and H is a nonzero closed U -invariant subspace, H contains a nonzero eigenvector for U . Indeed, if $0 \neq h_0 \in H$, then some pure-point spectral projection $P_\alpha h_0$ is nonzero; the projection may be obtained as the mean-ergodic limit

$$P_\alpha h_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \bar{\alpha}^n U^n h_0,$$

so $P_\alpha h_0 \in H$. Thus choose $0 \neq h \in H$ and $\alpha \in \mathbb{T}$ with $Uh = \alpha h$.

Now define a covariance map $R : H \rightarrow X$. On the dense subspace $j(X^*)$ set

$$Rj(x^*) = \int_X x \overline{x^*(x)} d\nu(x).$$

This is well defined and bounded, since

$$\|Rj(x^*)\| \leq \left(\int_X \|x\|^2 d\nu(x) \right)^{1/2} \|j(x^*)\|_{L^2(\nu)}.$$

We use the convention

$$\langle f, g \rangle_{L^2(\nu)} = \int_X f(x) \overline{g(x)} d\nu(x),$$

linear in the first variable. For every $y^* \in X^*$,

$$y^*(Rg) = \langle j(y^*), g \rangle_{L^2(\nu)} \quad (g \in H),$$

so R is injective on H : if $Rg = 0$, then g is orthogonal to $j(X^*)$, and therefore to all of H .

Finally, compute directly on the Koopman eigenvector h . For every $y^* \in X^*$,

$$\begin{aligned} y^*(ARh) &= (A^* y^*)(Rh) \\ &= \langle j(A^* y^*), h \rangle_{L^2(\nu)} \\ &= \langle Uj(y^*), h \rangle_{L^2(\nu)} \\ &= \langle j(y^*), U^{-1}h \rangle_{L^2(\nu)} \\ &= \langle j(y^*), \bar{\alpha}h \rangle_{L^2(\nu)} = \alpha \langle j(y^*), h \rangle_{L^2(\nu)} = y^*(\alpha Rh). \end{aligned}$$

Since the functionals in X^* separate points, $ARh = \alpha Rh$. Since R is injective and $h \neq 0$, the vector Rh is nonzero. Thus $\alpha \in \mathbb{T}$ is an eigenvalue of A . $\square$

Corollary 1. *Every 2-universal operator for invertible ergodic systems on a complex separable Banach space has a unimodular eigenvalue.*

Proof. Apply Theorem 1 to any nontrivial ergodic irrational rotation of the circle. Such a rotation is invertible and has pure point Koopman spectrum. The definition of 2-universality supplies an invariant full-support measure with finite second moment modeling it. $\square$

Corollary 2. *The Kalish-type operator on $C_0([0, 2\pi])$ appearing in Grivaux's Question 5.9 is not 2-universal for invertible ergodic systems.*

Proof. Grivaux records, citing Bayart–Grivaux, that this operator has no unimodular eigenvalue. This contradicts Corollary 1 if it were 2-universal. $\square$

Verification and Novelty Notes

The argument uses only the finite second moment in the definition of 2-universality, full support, and the pure point test system. It does not use Hilbert structure or cotype. This is weaker than Grivaux’s Hilbert-space Proposition 5.3, which obtains perfectly spanning unimodular eigenvectors for Hilbert-space 2-universal operators, but it applies to arbitrary complex separable Banach spaces and is enough to rule out no-eigenvalue examples.

Bounded novelty checks performed during the run:

- cheap run indexes for arXiv:1410.2957, universal hypercyclic operators, 2-universal operators, unimodular eigenvalues, and the Kalish-type $C_0([0, 2\pi])$ example;
- local arXiv-source searches for **2-universal**, universal operators, no-eigenvalue variants, Gaussian invariant measures, and Kalish-type operators;
- targeted web searches for exact phrases around Question 5.9, “universal operator no unimodular eigenvalue”, and “2-universal universal operators unimodular eigenvalue”.

These searches found Grivaux’s source paper, Bayart–Matheron’s related Gaussian-mixing discussion, and later survey/citation mentions, but no direct recorded answer to the 2-universal no-eigenvalue clause.

Limitations

This packet does not prove that a 2-universal operator has full-measure unimodular point spectrum, nor that its unimodular point spectrum is all of $\mathbb{T}$. Those stronger Hilbert-space questions are separate in Grivaux’s paper. It also does not decide whether there exists an ordinary universal operator, without finite second moment, admitting no unimodular eigenvalue; nor does it decide whether the Kalish-type C_0 operator is universal in that weaker sense.

Human Review Recommendation

Review the covariance map $R : H \rightarrow X$ and the final Koopman-eigenvector calculation showing that Rh is a genuine eigenvector for A . These are the only functional-analytic points where the finite-second hypothesis is used. If they check out, the application to the Kalish-type operator is immediate from the source’s no-eigenvalue statement.

References

- [1] S. Grivaux, *Some new examples of universal hypercyclic operators in the sense of Glasner and Weiss*, Trans. Amer. Math. Soc. 369 (2017), 7589–7629. arXiv:1410.2957.
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